Stopping really fast

11a

1 How well a car stops depends on a whole lot of things. Draw ticks in the table to show whether the listed feature helps a car to stop more quickly or whether the feature slows the stopping.

Feature	Assists faster stopping	Hinders faster stopping
Worn tyres		
Wide tyres		
Five people in the car		
Only the driver in the car		
Lots of luggage		
Very loud music		
Slow music		
Small car		
Heavy car		
Disc brakes		
Drum brakes		
Combined disc/drum brakes		
Big brake pads		
ABS brakes		
Tired driver		
Alert driver		
Distracted driver		
Low speed		
High speed		
Racing car driver		
Sunday driver		
Car with caravan		
Smooth bitumen road		
Dirt road		
Grassy road		
Wet road		
Road during a light shower		
Dry road		

How quickly a car stops depends on the reaction time of the driver (the time before the driver acts to slow the car) and on the braking time (when the driver is actually pushing down on the brake pedal).

2 Write a formula for calculating the stopping distance.

_____

Assume that the reaction time for a driver is one second. This reaction time would change very little for a particular driver and it doesn't depend on the driver's speed. However, what does change is how much ground is covered in that one second when the car is travelling at different speeds. The distance covered by a car in one second at various speeds becomes very important when considering stopping times.

3 Write the formula for speed. _____

4 Rearrange the speed formula and make distance its subject.

_____

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Remember when carrying out calculations that all units in your calculations must be matching units. Sometimes it is necessary to convert one unit to another to maintain consistency of units in calculations. Remember: To convert m/s to km/h, multiply the number of metres per second by 3.6 (1 m/s = 3.6 km/h). To change km/h to m/s, divide the kilometres per hour by 3.6 (1 km/h = 1/3.6 m/s or 0.28 m/s).

5 Calculate the reaction distances for a driver with a reaction time of 1 second for the speeds shown in the table below. Show your working out.

Speed (km/h)	Speed (m/s)	Time (s)	Reaction distance (m)
20		1	5.56
40		1	
50		1	
60		1	
80		1	
110		1	
200		1	

A braking test was carried out at a driver training school.

A driver drove a car to a certain speed and then pressed the brake pedal to stop the car as quickly as possible (without locking the wheels). At a low speed of 20 km/h, the car stopped within 3 metres. At 40 km/h, the braking distance was 10 metres. At 50 km/h, the braking distance was 15 m and at 60 km/h it was 22 m. At 80 km/h the car needed 38 m to come to a stop and at 100 km/h the braking distance had gone up to 60 m. At the NSW highway top speed of 110 km/h, the car took 72 m to stop.

6 Summarise the results of the braking test in the table at right.

7 Construct another table (below) to summarise the information gathered in this activity. The table should consider at least five different speeds and show the reaction distance, the braking distance and the total stopping distance for these five different speeds for a driver with reaction time of 1 second.

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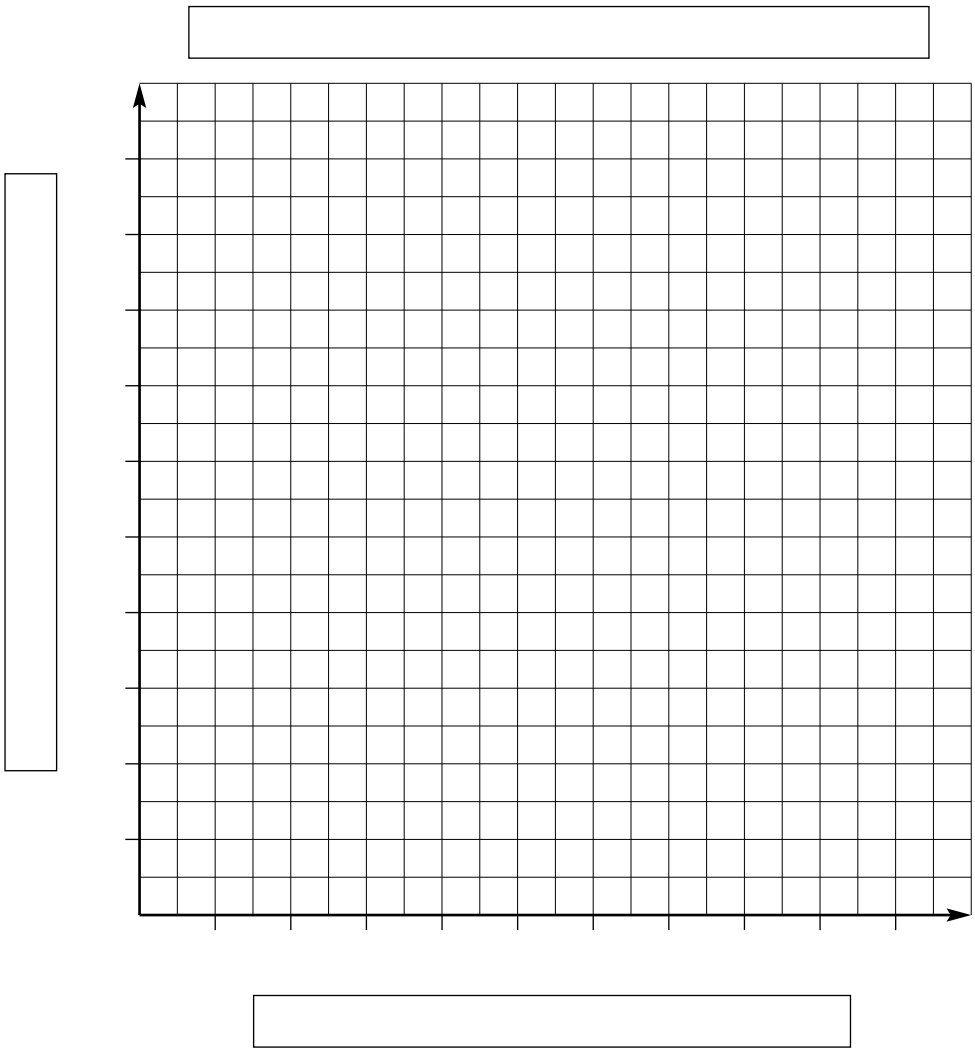
8 Use your completed table from question 7 to graph the reaction distance, braking distance and stopping distance for each speed. Plot speed on the horizontal axis and distance on the vertical axis. To highlight the different distances, use a red line for stopping distance, a blue or dotted line for reaction distance and a black or dashed line for braking distance. Label the axes and the curves. Give a title to your graph.



9 In considering reaction time and braking distance, everyone can formulate some personal driving advice. Complete the sentence below.

The _____ I drive the _____ distance I need to keep from the car in front and the _____ alert I must be.

10 Sometimes road signs are designed to alert drivers to changes in road conditions or possible dangers. Use the back of this sheet to draw the road signs that would be used for each of the following situations.

a approaching a school

b pedestrian crossing ahead

c approaching a railway crossing

d stock crossing ahead

e children's playground nearby

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